

References

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's Disease

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- [1] Iaccarino, H. F. et al. (2016). Gamma frequency entrainment attenuates amyloid load and modifies microglia. *Nature*, 540(7632), 230-235.
- [2] Martorell, A. J. et al. (2019). Multi-sensory gamma stimulation ameliorates Alzheimer's-associated pathology and improves cognition. *Cell*, 177(2), 256-271.
- [3] Tort, A. B. L. et al. (2010). Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2), 1195-1210.
- [4] Lawhern, V. J. et al. (2018). EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces. *Journal of Neural Engineering*, 15(5), 056013.
- [5] Lahijanian, M. et al. (2024). Auditory gamma-band entrainment enhances default mode network connectivity in dementia patients. *Scientific Reports*, 14, 13153.
- [6] Thompson, R. F. & Spencer, W. A. (1966). Habituation: A model phenomenon for the study of neuronal substrates of behavior. *Psychological Review*, 73(1), 16-43.
- [7] Chan, D. et al. (2025). Long-term safety and tolerability of gamma sensory stimulation for Alzheimer's disease. *Alzheimer's & Dementia*, 21(10), e70792.
- [8] Fortunato et al. (2023). Non-responder rates in auditory gamma entrainment studies. *Frontiers in Integrative Neuroscience*, 17.
- [9] Murdock, M. H. et al. (2024). Multisensory gamma stimulation promotes glymphatic clearance of amyloid. *Nature*, 627, 149-156.
- [10] Soula, M. et al. (2023). Forty-hertz light stimulation does not entrain native gamma oscillations in Alzheimer's disease model mice. *Nature Neuroscience*, 26, 570-578.
- [11] Meta AI. (2026). TRIBE V2: A Predictive Foundation Model for Brain Encoding. HuggingFace model card: facebook/tribev2.
- [12] Wilson, H. R. & Cowan, J. D. (1972). Excitatory and inhibitory interactions in localized populations of model neurons. *Biophysical Journal*, 12(1), 1-24.

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